

Milestone 3: Model Implementation 150 Points Possible

4/13/2026

Attempt 1



In Progress

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Unlimited Attempts Allowed

4/13/2026

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Overview

This project is a **comprehensive group assignment** that allows you to apply concepts learned in this course. You will work with **real data** to answer research questions, perform data mining techniques, and communicate findings through a professional project website.

The project consists of **four milestones** to ensure structured progress. Each milestone builds upon the previous one, and missing any prior requirement must be addressed before proceeding.

- **Project Format:** Hosted on a **GitHub repository** and a **public website**
- **Group Work:** Teams of **three students**
- **Final Deliverable:** Website + Report (8-12 pages in ACM format) + Source Code

Submission Guidelines

All project materials should be submitted via **Canvas** in the following format:

1. **Website URL** (must be publicly accessible)
2. **GitHub Repository URL** (including datasets, code, and reports)
3. **PDF Report** (full project documentation in ACM format)
4. **ZIP File** (containing source code and key results)
5. **Team Contribution Report** (detailing individual contributions)

Milestone 3: Model Implementation

Objective

In this milestone, students will apply machine learning models to analyze their datasets and answer their research questions. They will implement multiple models, evaluate their performance, and fine-tune them for better results.

Requirements

Create a Website Tab "Models Implemented" which showcases below:

- **Models Implemented**

Each team must implement **at least four machine-learning models** from different categories:

- **Frequent Pattern Mining** (e.g., Apriori, FP-Growth)
- **Classification** (e.g., Decision Trees, SVM, k-NN, Naïve Bayes)
- **Clustering** (e.g., K-Means, DBSCAN, Hierarchical Clustering)
- **Regression** (e.g., Linear Regression, Logistic Regression)
- **For each model, include:**
 - **Why the model was chosen** (justify selection based on dataset characteristics).
 - **Model assumptions** (e.g., SVM assumes data is linearly separable).
 - **Hyperparameter tuning** (adjustments to improve model performance).
 - **Challenges faced & solutions** (if applicable).

- **Performance Evaluation**

- **Each team must** Select appropriate **evaluation metrics** based on model type:
 - **Classification:** Accuracy, Precision, Recall, F1-score, ROC-AUC
 - **Regression:** RMSE, MSE, R²-score
 - **Clustering:** Silhouette Score, Davies-Bouldin Index
 - **Frequent Pattern Mining:** Support, Confidence, Lift
- **Compare model performances** and explain which approach worked best.
- Justify why one model performs better than another for their specific dataset.

- **Data Formatting for Models**

- Ensure that **dataset transformations align with model requirements:**
 - **SVM** requires numerical data (convert categorical features).
 - **Decision Trees** can handle categorical variables.
 - **Clustering algorithms** may require feature scaling.
- Include **before-and-after data transformation snapshots** to show the preprocessing steps.

Submission Requirements

- **Website URL** (updated with the "Models Implemented" tab)
- **GitHub URL** (including model scripts and datasets)
- **PDF Report** (detailing model selection, training process, evaluation, and insights)
- **Poster Presentation Slide (version 1)**

✓ **View Rubric**

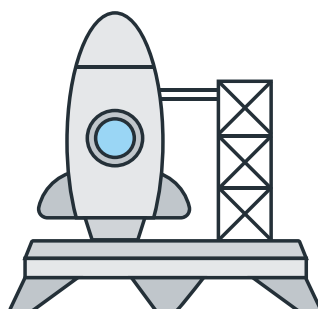
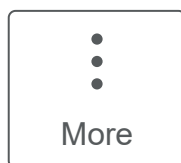
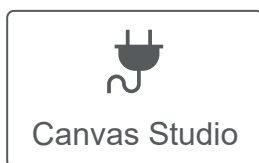
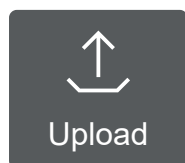
Some Rubric (8)

Criteria	Ratings	Pts
Previous Updates view longer description	10 pts Full Marks	0 pts No Marks / 10 pts

Some Rubric (8)

Criteria	Ratings	Pts
Modelling view longer description	120 pts Full Marks	0 pts No Marks / 120 pts
Report view longer description	10 pts Full Marks	0 pts No Marks / 10 pts
Submission view longer description	10 pts Full Marks	0 pts No Marks / 10 pts
		Total Points: 0

! Keep in mind, this submission will count for everyone in your Project Groups group.

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